# Supervised Neural Network Project: GNN-Based BERT for Understanding Context from Music

Course: Neural Networks (CSE425 / EEE474 / CSE715)

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Submission Deadline: **2nd October, 2026**

## 1 Project Motivation

Music is a **multi-layered signal** where "context" spans melody, harmony, rhythm, lyrics, metadata tags, and listener-described semantics. A single clip may simultaneously express:

- **Genre and era** (e.g., Jazz, 1960s)
- **Mood and emotion** (e.g., melancholic, high arousal)
- **Harmonic structure** (chord progressions, key changes)
- **Lyrical semantics** (themes, sentiment in vocals)
- **Timbral texture** (instrumentation, production style)

Pure sequence models (CNN/RNN on spectrograms) capture local patterns but often miss **relational structure**: how chord $C \to G \to Am$ relates to a lyrical theme, or how repeated segments form a song graph.

**Goal:** Build a hybrid **BERT + Graph Neural Network (GNN)** system that understands musical context by combining:

1. **BERT** — contextual language representations from lyrics, tags, and natural-language music descriptions (MusicCaps).
2. **GNN** — message passing on music structure graphs (chord transitions, segment similarity, co-occurrence of audio events).

Unlike generative music projects, this assignment focuses on **understanding and prediction**: multi-label tagging, emotion regression, and cross-modal alignment between audio structure and text.

## 2 Problem Definition

Let a music track be represented as a tuple:

$$\mathcal{T} = (\mathbf{X}_{\text{audio}},\ \mathbf{X}_{\text{text}},\ \mathbf{G},\ \mathbf{y}),$$

where:

- $\mathbf{X}_{\text{audio}}$ = log-mel spectrogram or chroma features over $T$ frames,
- $\mathbf{X}_{\text{text}}$ = tokenized lyrics, user tags, or caption tokens,

- $\mathbf{G} = (\mathcal{V}, \mathcal{E})$ = music structure graph (nodes = segments/chords/events; edges = transitions or similarity),
- $\mathbf{y}$ = context labels (genre, mood tags, valence/arousal, etc.).

**BERT text encoder** (frozen or fine-tuned) maps text to contextual embeddings:

$$\mathbf{H}_{\text{text}} = \text{BERT}(\mathbf{X}_{\text{text}}) \in \mathbb{R}^{L \times d}.$$

**GNN encoder** with $L$ layers updates node features via message passing:

$$\mathbf{h}_i^{(l+1)} = \sigma\Big(\mathbf{W}^{(l)} \cdot \text{AGG}\big(\mathbf{h}_i^{(l)},\ \{\mathbf{h}_j^{(l)} : j \in \mathcal{N}(i)\}\big)\Big),$$

where $\mathbf{h}_i^{(0)}$ is initialized from audio segment embeddings or chord features.

**Fusion readout** combines graph-level representation $\mathbf{g}$ and text CLS vector $\mathbf{t}$:

$$\mathbf{z} = \text{Fusion}(\mathbf{g}, \mathbf{t}), \qquad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{z} + \mathbf{b}).$$

**Objective** (multi-label context understanding):

$$\mathcal{L} = -\sum_{k=1}^{K} \Big[ y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \Big] + \lambda \, \mathcal{L}_{\text{aux}}.$$

# 3 Dataset Requirements

Students must use **at least one primary audio dataset** and **one text/tag dataset** from Table 1. Advanced tasks should combine **FMA or MagnaTagATune** with **MusicCaps** or **DEAM**.

### Recommended primary pairing

- **Medium/Hard:** `FMA-medium` (audio + genre/tags) + chord/segment graphs built from chroma features.
- **Advanced:** `MusicCaps` captions (BERT) aligned with audio graphs from corresponding 10 s clips.
- **Emotion extension:** `DEAM` for valence/arousal regression as auxiliary $\mathcal{L}_{\text{aux}}$.

### Preprocessing pipeline

1. **Audio:** Resample to 22,050 Hz; extract log-mel spectrogram (128 bins) or chroma (12 bins); normalize per track.
2. **Segmentation:** Split each track into fixed windows (e.g., 5–10 s) or beat-synchronous segments using librosa.
3. **Graph construction:**
    - *Chord-transition graph:* nodes = unique chords; edges = observed transitions weighted by count.

Table 1: Recommended datasets for GNN–BERT music context understanding.

| Dataset Name | Context / Label Coverage | Task | Official Link |
|---|---|---|---|
| FMA (small / medium) | Genre (8–16 classes), top tags, artist & album metadata | 1–3 | `FMA Download` |
| MagnaTagATune | 188 multi-label tags (genre, mood, instrument); 25,877 clips | 1, 3 | `MagnaTagATune` |
| GTZAN | 10 genres; 1,000 tracks (30 s clips); standard Easy baseline | 1–2 | `GTZAN Download` |
| DEAM | Continuous **valence** & **arousal** (1–9) every 0.5 s | 3–4 | `DEAM Download` |
| MusicCaps | 5,521 clips with expert natural-language captions (Google) | 1, 4 | `MusicCaps` |
| Million Song Dataset + tagtraum | Last.fm-style tags for 188,916 tracks (Echo Nest audio features) | 1, 3 | `MSD Download` |
| Lakh MIDI Clean | Symbolic MIDI for chord-transition & melody **graph** construction | 2–3 | `LMD Download` |
| EmoMusic | 1,000 clips with quadrant emotion labels (happy/sad × calm/energetic) | 3 | `EmoMusic` |

**Task key:** 1 = BERT tags — 2 = GNN structure — 3 = GNN–BERT fusion — 4 = contrastive retrieval. Use $\geq 1$ audio dataset and $\geq 1$ text/tag source.

- *Segment graph:* nodes = time segments; edges = temporal adjacency + cosine similarity of MFCC/chroma $> \tau$.

4. **Text:** Tokenize lyrics/tags/captions with BERT tokenizer (max length 128–256); pad/truncate.
5. **Splits:** Use official FMA / MagnaTagATune splits; for DEAM use standard train/val partition; **no artist leakage** across train/test when possible.

# 4 Model Tasks and Mathematical Formulation

## 4.1 Task 1 (Easy): BERT Baseline for Music Tag Understanding

**Goal.** Implement a **BERT-based multi-label classifier** on textual music context (tags, captions, or lyrics) without graph structure.

**Model.**

$$\mathbf{t} = \mathrm{BERT}_{\mathrm{CLS}}(\mathbf{X}_{\mathrm{text}}), \qquad \hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t} + b_k).$$

**Loss (binary cross-entropy per tag $k$):**

$$\mathcal{L}_{\mathrm{BERT}} = -\frac{1}{K}\sum_{k=1}^{K}\big[y_k \log \hat{y}_k + (1-y_k)\log(1-\hat{y}_k)\big].$$

**Deliverables.**

- BERT fine-tuning code (HuggingFace `bert-base-uncased` or `distilbert-base-uncased`)
- Results on MagnaTagATune tag subset (top-50 tags) or MusicCaps caption → tag proxy task
- Macro-F1 / Micro-F1 curves vs. training epochs
- 5 example predictions with attention visualization (optional)

### 4.2 Task 2 (Medium): GNN on Music Structure Graphs

**Goal.** Build a **GraphSAGE or GAT** encoder on segment/chord graphs using **audio-only** node features; predict genre or top tags.

**GraphSAGE update:**

$$\mathbf{h}_i^{(l+1)} = \sigma\Big(\mathbf{W}^{(l)} \cdot \mathrm{CONCAT}\big(\mathbf{h}_i^{(l)},\ \mathrm{MEAN}_{j\in\mathcal{N}(i)}\mathbf{h}_j^{(l)}\big)\Big).$$

**Graph readout** (mean pooling):

$$\mathbf{g} = \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} \mathbf{h}_i^{(L)}, \qquad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{g} + \mathbf{b}).$$

**Deliverables.**

- Graph construction scripts (chroma/MFCC segment graphs)
- GNN implementation (PyTorch Geometric)
- Genre classification on GTZAN or FMA-small
- Comparison vs. CNN baseline on mel-spectrogram

### 4.3 Task 3 (Hard): GNN–BERT Fusion for Multi-Context Understanding

**Goal.** Fuse **structural** (GNN) and **semantic text** (BERT) representations to predict **multi-label context**: genre + mood tags + (optional) valence/arousal.

**Cross-attention fusion (recommended):**

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{Q} = \mathbf{g}\mathbf{W}_Q,\ \mathbf{K} = \mathbf{H}_{\text{text}}\mathbf{W}_K,$$

$$\mathbf{z} = \mathrm{CONCAT}(\mathbf{g},\ \mathbf{A}\mathbf{H}_{\text{text}}), \qquad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{z}).$$

**Multi-task loss:**

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha\,\|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + \beta\,\|\mathbf{a} - \hat{\mathbf{a}}\|_2^2,$$

where $(\mathbf{v}, \mathbf{a})$ are DEAM valence/arousal targets when available.

**Deliverables.**

- End-to-end GNN–BERT fusion model
- Ablation: BERT-only, GNN-only, early concat, cross-attention
- Results on FMA-medium or MagnaTagATune (Macro-F1, AUC-PR)
- t-SNE of $\mathbf{z}$ coloured by genre and mood
- 3 case studies showing graph paths + caption/lyric alignment

### 4.4 Task 4 (Advanced): Cross-Modal MusicCaps Alignment

**Goal.** Learn a **shared embedding space** between audio graphs and natural-language descriptions (MusicCaps) using contrastive learning.

**InfoNCE contrastive loss** for paired (graph, caption) $(\mathbf{g}_i, \mathbf{t}_i)$:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_j)/\tau)},$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v}/\|\mathbf{u}\|\|\mathbf{v}\|$ and $\tau$ is temperature.

**Retrieval metrics:** Caption → Audio R@1, R@5, R@10; Audio → Caption R@K.

**Deliverables.**

- Dual-encoder GNN–BERT with contrastive training
- Retrieval evaluation table on MusicCaps test split
- 10 qualitative retrieval examples (query caption → top-3 matched clips)
- Zero-shot tag prediction from captions vs. Task 3 supervised model

## 5 Task Roadmap Summary

Table 2: Four-task roadmap with objectives and deliverables.

| Task | Model | Key equations | Deliverables |
|---|---|---|---|
| 1 (Easy) | BERT tag classifier | $\hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t})$; $\mathcal{L}_{\text{BERT}}$ | BERT code; tag F1; 5 predictions |
| 2 (Medium) | GNN on segment/chord graph | GraphSAGE update; mean readout $\mathbf{g}$ | Graph builder; GNN code; vs. CNN |
| 3 (Hard) | GNN–BERT fusion | Cross-attention; multi-task $\mathcal{L}$ | Ablations; t-SNE; case studies |
| 4 (Advanced) | Contrastive dual-encoder | InfoNCE $\mathcal{L}_{\text{NCE}}$; R@K | Retrieval table; 10 qual. examples |

## 6 Evaluation Metrics

**Tag / genre classification.**

$$\text{Prec}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Rec}_k = \frac{TP_k}{TP_k + FN_k}, \quad \text{F1}_k = \frac{2\,\text{Prec}_k\,\text{Rec}_k}{\text{Prec}_k + \text{Rec}_k}.$$

**Macro-F1** $= \frac{1}{K}\sum_k \text{F1}_k$; **Micro-F1** pools globally.

**AUC-PR.** Area under precision–recall curve per tag; report mean AUC-PR over tags.

**Emotion regression (DEAM).**

$$\text{MAE}_v = \frac{1}{N}\sum_{i=1}^{N} |v_i - \hat{v}_i|, \qquad R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

**Graph coherence score (optional analysis).** Measure whether high-attention edges align with repeated chord patterns:

$$S_{\text{graph}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \mathbb{1}[\cos(\mathbf{h}_i, \mathbf{h}_j) > \tau].$$

**Human evaluation (Task 4).** Minimum 5 listeners rate whether retrieved clip matches caption on scale $[1, 5]$.

# 7 Algorithms for the Tasks

**Algorithm 1** Task 1: BERT Multi-Label Tag Classifier

**Require:** Tag corpus $\mathcal{D} = \{(\mathbf{X}_{\text{text}}^{(i)}, \mathbf{y}^{(i)})\}$
**Require:** Pretrained BERT, learning rate $\eta$, epochs $E$
Add classification head $\mathbf{W}, \mathbf{b}$ on top of CLS token
**for** epoch $= 1$ to $E$ **do**
  **for** each mini-batch in $\mathcal{D}$ **do**
    $\mathbf{t} \leftarrow \text{BERT}_{\text{CLS}}(\mathbf{X}_{\text{text}})$
    $\hat{\mathbf{y}} \leftarrow \sigma(\mathbf{W}\mathbf{t} + \mathbf{b})$
    $\mathcal{L} \leftarrow \text{BCE}(\mathbf{y}, \hat{\mathbf{y}})$
    Update BERT (optional) and head via $\eta\nabla\mathcal{L}$
  **end for**
**end for**
**return** fine-tuned BERT tag model

**Algorithm 2** Task 2: GNN Encoder on Music Segment Graph

**Require:** Audio tracks $\{\mathcal{T}_i\}$, label set $\mathbf{y}$
**for** each track $\mathcal{T}$ **do**
  Extract segment features $\mathbf{h}_i^{(0)}$ from mel/chroma windows
  Build graph $\mathbf{G} = (\mathcal{V}, \mathcal{E})$ (temporal + similarity edges)
**end for**
**for** epoch $= 1$ to $E$ **do**
  **for** each graph $\mathbf{G}$ in batch **do**
    **for** layer $l = 0$ to $L - 1$ **do**
      $\mathbf{h}_i^{(l+1)} \leftarrow \text{GraphSAGE}(\mathbf{h}_i^{(l)}, \mathcal{N}(i))$
    **end for**
    $\mathbf{g} \leftarrow \text{MEANPOOL}(\{\mathbf{h}_i^{(L)}\})$
    $\mathcal{L} \leftarrow \text{BCE}(\mathbf{y}, \sigma(\mathbf{W}\mathbf{g}))$
    Update GNN parameters
  **end for**
**end for**

# 8 Baseline Models for Comparison

Students must compare against **at least two baselines**:

- **B1:** Majority-class / random tag predictor

**Algorithm 3** Task 3: GNN–BERT Fusion for Context Understanding

**Require:** Paired data $(\mathbf{G}, \mathbf{X}_{\text{text}}, \mathbf{y})$

$\mathbf{H}_{\text{text}} \leftarrow \text{BERT}(\mathbf{X}_{\text{text}}), \quad \mathbf{t} \leftarrow \mathbf{H}_{\text{text}}[\text{CLS}]$

$\mathbf{g} \leftarrow \text{GNN_Readout}(\mathbf{G})$

$\mathbf{z} \leftarrow \text{CrossAttention}(\mathbf{g}, \mathbf{H}_{\text{text}})$ ▷ or $\text{CONCAT}(\mathbf{g}, \mathbf{t})$

$\hat{\mathbf{y}} \leftarrow \sigma(\mathbf{W}\mathbf{z} + \mathbf{b})$

$\mathcal{L} \leftarrow \mathcal{L}_{\text{tags}} + \alpha \mathcal{L}_{\text{emotion}}$

Backprop through fusion, GNN, and (partial) BERT

**Algorithm 4** Task 4: Contrastive GNN–BERT (MusicCaps)

**Require:** Paired $(\mathbf{G}_i, \text{caption}_i)$, temperature $\tau$, batch size $N$

$\mathbf{g}_i \leftarrow \text{Normalize}(\text{GNN}(\mathbf{G}_i))$

$\mathbf{t}_i \leftarrow \text{Normalize}(\text{BERT}_{\text{CLS}}(\text{caption}_i))$

**for** each training step **do**

    Compute similarity matrix $\mathbf{S}_{ij} = \mathbf{g}_i^\top \mathbf{t}_j / \tau$

    $\mathcal{L}_{\text{NCE}} \leftarrow -\frac{1}{N} \sum_i \log \frac{\exp(\mathbf{S}_{ii})}{\sum_j \exp(\mathbf{S}_{ij})}$

    Update encoders to minimize $\mathcal{L}_{\text{NCE}}$

**end for**

Evaluate retrieval R@1, R@5, R@10 on held-out pairs

- **B2:** CNN on mel-spectrogram (no graph, no text)
- **B3:** BERT-only (Task 1)
- **B4 (optional):** PCA + MLP on hand-crafted audio features

Table 3: Illustrative performance comparison (replace with your experimental results).

| Model | Macro-F1 | AUC-PR | MAE (emotion) | R@5 (retrieval) |
|---|---|---|---|---|
| Random tags | 0.05 | 0.12 | – | 0.02 |
| CNN mel-spec | 0.41 | 0.38 | 1.25 | – |
| Task 1: BERT-only | 0.48 | 0.44 | – | – |
| Task 2: GNN-only | 0.52 | 0.47 | 1.10 | – |
| Task 3: GNN–BERT | 0.61 | 0.55 | 0.92 | – |
| Task 4: Contrastive | 0.55 | 0.50 | – | 0.38 |

## 9 Judging Rubric

### Marks distribution (100 total)

| Component | Marks | Notes |
|---|---|---|
| Task 1 (Easy) | 18 | BERT tags + F1 report |
| Task 2 (Medium) | 22 | GNN graph + CNN baseline |
| Task 3 (Hard) | 22 | Fusion + ablations |
| Task 4 (Advanced) | 18 | Contrastive retrieval (optional → bonus) |
| Report + GitHub | 20 | Reproducibility + paper quality |
| **Total** | **100** | |

Table 4: Final grading rubric (maps to 15% course grade; total 100 marks).

| Category | Criteria | Weight |
|---|---|---|
| Dataset & preprocessing | Correct splits, no leakage, graph construction documented | 15% |
| Model implementation | BERT + GNN correct, stable training, reproducible code | 25% |
| Context understanding quality | Strong tag/emotion/retrieval results vs. baselines | 20% |
| Baseline comparison | $\geq 2$ baselines with fair experimental setup | 15% |
| Metrics & analysis | Macro-F1, AUC-PR, ablations, visualizations | 15% |
| Report & presentation | Clear NeurIPS/IEEE-style report, diagrams, discussion | 10% |

## 10 Final Submission Requirements

Each group must submit:

1. **GitHub repository** or ZIP with full source code
2. **Preprocessed graph samples** (at least 20 example `.pt` / `.json` graphs)
3. **Evaluation tables + plots** (F1, AUC-PR, t-SNE, retrieval examples)
4. **Final report PDF** (6–10 pages; NeurIPS / IEEE / ICML Overleaf template)
5. **Demo notebook** (`notebooks/demo_context.ipynb`) with one end-to-end inference example

### GitHub project structure

```
gnn-bert-music-context/
  README.md
  requirements.txt
  config.yaml
  data/
    raw/                  # FMA, MagnaTagATune, MusicCaps downloads
    processed/            # graphs, mel-spec, BERT caches
    splits/               # train/val/test JSON
  notebooks/
    eda.ipynb
    demo_context.ipynb
  src/
    audio_features.py     # mel, chroma, segmentation
    graph_builder.py      # chord + segment graphs
    bert_encoder.py
    gnn_model.py          # GraphSAGE / GAT
    fusion_model.py       # cross-attention GNN-BERT
    contrastive.py        # Task 4 InfoNCE
    train.py
    evaluate.py
  results/
    metrics.json
```

```
  plots/
  retrieval_examples/
report/
  final_report.pdf
```

### Report template links (Overleaf)

- NeurIPS 2024: `https://www.overleaf.com/latex/templates/neurips-2024/tpsbbrdqcmsh`
- IEEE Conference: `https://www.overleaf.com/latex/templates/ieee-conference-template`
- ICML 2025: `https://www.overleaf.com/latex/templates/icml2025-template`